A Two-Shell Membrane Model for BoxOfActin

Fluid lipid bilayer (outer) + actin cortex (inner) — design sketch, rooted in cell biology

BoxOfActin DTS membrane v2 · 2026-06-23 · branch `membrane-render-and-dense-v5`

> **One sentence.** Split the single DTS membrane into two coupled shells — a compliant **fluid bilayer** that actin *pushes* and signals through, and a stiffer **actin cortex** that actin *anchors* to and draws its reaction from — because the cell does exactly this, and because conflating the two is the direct cause of the non-physical "superman" free-ride we keep hitting.

1 · Why one shell fails — the mechanical argument

In the current single-shell model a formin-nucleated filament both **pushes** the membrane (steric, via its tip-spheres) *and* is **tethered** to that same membrane (the formin–membrane bond). The membrane is therefore the filament's *own* reaction anchor. In overdamped dynamics that is unstable in a specific way:

- Pushing alone would send the filament *inward* (equal-and-opposite recoil) — harmless.
- But the bond's rest target tracks the very vertex the filament pushes *out*, so the bond hauls the filament outward after it. With no *external* reaction, filament + membrane inflate together — measured at ~0.12 µm of co-drift over a 12k-step dendritic run (actin 95th-percentile radius and membrane radius rising in lockstep). This is the "superman" ride-out.
- Removing the coupling kinematically does not work: an instantly-tracking anchor rides out; a lagging anchor builds unrelieved bond strain and **blows up** (a vertex flung to r≈6.6 µm in test). The two failure modes *bracket* the problem and show that the fix must be a **physical reaction surface with its own dynamics**, not an anchor trick.

That surface, in a real cell, is not the lipid bilayer. It is the actin cortex.

2 · The biology: two distinct structures, two distinct jobs

The cell surface is a **composite**: a fluid lipid bilayer floating on, and tethered to, a thin load-bearing actin cortex [1,2,5]. They have different compositions, thicknesses, stiffnesses, and dynamics — and they do different mechanical jobs. The single DTS shell has been trying to be both.

Property	Lipid bilayer (plasma membrane)	Actin cortex	Maps to
What it is	~4–5 nm fluid phospholipid bilayer + embedded/peripheral proteins	Cross-linked F-actin meshwork + myosin II, just under the membrane	—
Thickness	~4–5 nm	~100–400 nm (measured 116–143 nm); mesh 50–200 nm [3,4]	shell separation
In-plane mechanics	2D **fluid**: no shear resistance, lipids diffuse freely; total area nearly	2D **solid/gel**: resists shear, cross-linked; remodels by turnover	outer = flips/remesh;

	inextensible ($K_A \approx 0.2$–0.3 N/m)		inner = stiffer
Bending / stiffness	Low bending rigidity $\kappa \approx 10$–$25\ k_BT$ (~ 0.4–1.0×10^{-19} J)	Much stiffer in bulk; bears the mechanical load and sets cell shape [5]	$\kappa_{inner} \gg \kappa_{outer}$
Tension	Membrane tension $\approx 3\times10^{-6} - 3\times10^{-4}$ N/m; buffered by reservoirs (folds, caveolae)	Cortical tension $\approx \zeta\Delta\mu \cdot h/2$ (myosin active stress × thickness); typically 10^{-4}–10^{-3} N/m, often dominant [3,6]	tension knobs per shell
Dynamics	Fast lipid diffusion; passively follows; flows to relieve local tension	Turns over on ~seconds–minutes; actively contractile (myosin); generates pressure	outer fast, inner slow
Mechanical role	Barrier; **signaling platform**; steric surface; deforms (blebs, ruffles)	**Load-bearing**; reaction surface for polymerization; shape; contractility	see §3

3 · Functional assignment in the model

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   OUTSIDE
 ~~~~~~~~~~~~~~~~~~~~~~~~~~~~~   outer shell = LIPID BILAYER
  |    |    |    |    |    |     fluid, compliant: flips + split/collapse
  o    o    o    o    o    o     MCA linkers: ERM-like
 ===========================     (sliding, force-limited, breakable)
 /|\   /|\       \|/   /|\       inner shell = ACTIN CORTEX
  |     |         |     |        stiff, slow-remodel = reaction surface
  |     |         |     |
  |   explicit formin / Arp2/3 filaments:
  |     - ANCHOR to the cortex   (reaction, bond)
  |     - PUSH the bilayer       (steric tip-spheres, no bond)
   CYTOPLASM (interior)
```

Outer shell — the lipid bilayer

- **Compliant signaling & steric surface.** This is the surface that locates/organizes signals and coarsely gauges response (bleb vs. protrusion) — the role you actually want from it.
- **Fluid:** edge flips + split/collapse give in-plane redistribution (area streams; no shear stiffness).
- **Pushed, never bonded.** Actin tip-spheres exert steric force on it (stall-capped); it deforms and tracks growth but is *not* an anchor — so it cannot drag actin.
- Already built: Jülicher bending, area/volume constraints, flips, split, collapse, end-sphere steric.

Inner shell — the actin cortex

- **The reaction surface.** Formin/Arp2/3 anchors bond *here*. Being stiff, it resists fast ride-out; having its own overdamped dynamics, it relieves strain by moving slowly (no blow-up) — the property the kinematic anchor lacked.
- **Stiffer + slower:** higher κ and K_A, heavier vertices (more drag), slower/less-frequent remesh; optionally myosin-like contractility later (cortical tension).

- **Coarse-grained.** It stands in for the dense cross-linked meshwork (mesh 50–200 nm) — not every cortical filament, but the load-bearing continuum the active protrusive filaments push off of.

> **Design decision A — what *is* the cortex, explicitly?** Two readings: (a) the inner shell is a coarse continuum cortex and the explicit dendritic filaments are *active protrusive* actin that anchors into it (recommended — clean separation of "bulk load-bearing" vs "active pushing"); or (b) the explicit filament network *is* the cortex and the inner shell is only the cell-body boundary they nucleate from. (a) matches the biology of a persistent cortex feeding transient protrusions and is the simpler mechanical story.

4 · Coupling the two shells — the membrane–cortex attachment (MCA)

This is the crux, and the biology is unusually specific and helpful. The bilayer is held to the cortex by discrete linker proteins — chiefly the **ERM family (ezrin, radixin, moesin)** and myosin-I — whose FERM domain binds membrane PI(4,5)P_2 and whose tail binds F-actin [1,2]. Their measured behavior dictates the coupling model:

- **Force-limited & transient.** A single ezrin lasts ~0.1 s and bears ~0.5–4 pN; small assemblies sustain ~4 pN for seconds–minutes [7]. The link is a *population* of weak bonds, not a weld.
- **Sliding anchor.** Linker complexes *slide along F-actin, relaxing force parallel to the surface to ~0* while still bearing perpendicular load [7]. So the bilayer can **flow tangentially** over the cortex while staying attached normally — exactly the in-plane fluidity we want, with normal containment.
- **Breakable → blebs.** When local load exceeds attachment (high cortical tension / myosin pressure), the membrane **detaches** and a bleb herniates out under cytoplasmic hydrostatic pressure [3,4,6]. Detachment is not a bug — it is a modeled cellular event.

That gives a natural spectrum of coupling models, from crude to faithful:

Option	Mechanism	Biological basis	Behavior / cost
A. Rigid co-move	Outer vertices slaved to nearest inner vertex (or shared)	None (limit of infinitely strong MCA)	Trivial; kills superman but also kills bleb/ruffle and bilayer fluidity. Baseline only.
B. Normal confinement	Soft one-sided springs keeping the gap near a rest separation (cortex thickness); no tangential force	MCA holds the membrane ~tens of nm off the cortex	Bilayer free to flow in-plane; cortex sets the radius. Simple; no explicit linkers.
C. Sliding linkers ERM	Discrete tethers, inner↔outer, that bear **normal** load but relax **tangential** force (re-select the attachment point as the mesh flows)	Ezrin sliding-anchor [7]	Bilayer flows over a held cortex; normal coupling without tangential drag. Reuses our `BOA_SLIDE_ANCHOR` + PAIRS drag-weighted bond.
D. Breakable sliding linkers ERM + bleb	As C, plus a per-linker detachment force; broken patches lose normal coupling	MCA rupture under load → bleb nucleation [3,6]	Everything in C *plus* emergent blebbing where load > attachment. The faithful model.

E. + cytoplasmic pressure	D, plus an outward hydrostatic pressure term inside (volume/osmotic)	Myosin-driven cytoplasmic pressure inflates detached membrane	Lets a detached patch actually balloon into a bleb. Add only when blebs are the target.

> **Recommendation.** Build toward **C → D**: **sliding, force-limited linkers** as the MCA. C alone already (i) fixes superman (reaction goes to the stiff cortex, not the pushed bilayer), (ii) preserves bilayer in-plane fluidity (tangential force relaxes to ~0 — the sliding anchor), and (iii) keeps the bilayer near the cortex (normal load borne). Adding a detachment force (D) then yields **blebbing for free**, which is one of the emergent behaviors you named as a goal. Start at C; promote to D/E when blebs are wanted. Options A/B are useful sanity baselines on the way.

5 · Why this resolves "superman" by construction

- **Push and reaction live on different surfaces.** Actin's anchor (reaction) is on the stiff cortex; its push (deformation) is on the fluid bilayer, which it does not bond to. The feedback loop that inflated both together is severed.
- **The cortex resists fast, yields slow.** As a physical overdamped shell it bears the anchor load on the step timescale (no ride-out) but can remodel under sustained force (no strain blow-up) — precisely the "stiff-but-yielding" reaction surface the bracketing experiments demanded.
- **Tangential flow is allowed, normal containment kept.** The sliding linker relaxes in-plane force (so the bilayer flows and the anchor does not surf the mesh) while bearing normal load (so the bilayer stays on the cortex). This is also the cleaner cure for the older anchor-surfing/drift artifacts.

6 · Implementation sketch

Most pieces already exist; this is mostly wiring two instances together.

1. **Two `Membrane` instances** (the `theMembranes` list is already multi-instance): an outer bilayer (R≈1.2 µm, current parameters) and an inner cortex (R≈1.1 µm — the ~100 nm gap is the cortex thickness), with stiffer κ/K_A, heavier vertex drag, and slower remesh cadence.
2. **Route the anchor:** formin/Arp2/3 bonds target the nearest *cortex* vertex, not the bilayer (one-line change in `collideActin`'s anchor selection).
3. **Route the push:** actin end-spheres steric-push the *bilayer* only (already the default surface); the cortex need not collide with actin, or collides weakly.
4. **MCA linkers (the new piece):** a set of inner↔outer tethers built from the existing PAIRS drag-weighted bond + sliding re-selection (`BOA_SLIDE_ANCHOR`), bearing normal load and relaxing tangential; later add a per-linker break force for blebs (option D).
5. **Reuse:** bilayer fluidity (flips/split/collapse) and end-sphere steric are done; the cortex uses the same DTS machinery with stiffer parameters and slower remeshing.

Quantity	Biological value	Model handle (proposed)

Shell separation (cortex thickness)	~100–150 nm	$R_{outer} - R_{inner} \approx 0.1$ µm
Bilayer bending κ	10–25 $k_B T$	`dtsKappaBend` ≈ 1e-19 J (current)
Cortex stiffness	≫ bilayer (load-bearing)	higher κ/K_A, ↑ vertex drag, slow remesh
MCA linker force	~4 pN per ERM complex; population	PAIRS bond cap per linker (≈ few pN)
MCA tangential	relaxes to ~0 (sliding)	sliding re-selection (`BOA_SLIDE_ANCHOR`)
Bleb threshold	MCA rupture under cortical tension	per-linker break force (option D)

7 · Open decisions before building

- **Cortex identity** (Design Decision A above): coarse continuum cortex + active filaments (recommended), or filament-network-as-cortex.
- **Coupling rung:** start at C (sliding linkers) — agreed? Or first stand up A/B as a scaffold to verify the two-shell plumbing before adding linker physics.
- **Does actin also sterically clear the cortex,** or only anchor to it and push the bilayer? (Biology: actin *is* the cortex region, so likely anchor-only, no self-steric.)
- **Cortex remodeling:** does the inner shell turn over / flow (cortex treadmills on ~min), or is it quasi-static for now? Quasi-static is fine for a first cut.
- **Cytoplasmic pressure** (option E): defer until blebbing is the explicit target.

References

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BoxOfActin · two-shell DTS membrane design sketch · prepared for review before implementation. Companion to `MEMBRANE_DTS_DESIGN.md` and the 2026-06-22/23 JOURNAL entries (end-sphere actin push, remesher, superman diagnosis).